Milestone 1 (M1) Report

Design and Implementation of an OFDM-Based Communication and Sensing Framework Using GNU Radio

Problem Formulation, Literature Review, State-of-the-Art Positioning, and Timeline

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Project ID	G17, g17_gsoc_isac_ind
Course	ECE 310, Open-Source Contribution Track (GNU Radio / GSoC, ISAC domain)
Target Organization	GNU Radio (Google Summer of Code)
Domain	Integrated Sensing and Communication (ISAC), OFDM waveform
Repository	https://github.com/vamsip-cell/g17_gsoc_isac_ind.git

1. Problem Statement

1.1 Statement

GNU Radio already supports OFDM communication well, but it has no way to reuse that same signal for sensing: estimating the range and speed of whatever it reflects off. Every practical ISAC system needs one waveform to do both jobs, without doubling the hardware or the spectrum it uses. This project addresses that gap directly. GNU Radio's OFDM receiver already computes information that could support sensing, but nothing today exposes or reuses it for that purpose.

1.2 Motivation and Significance

ISAC matters because it removes the need for separate communication and radar hardware, which is relevant for 6G, vehicle-to-everything communication, and other next-generation systems. OFDM is a natural starting point since it already underlies 4G, 5G, and Wi-Fi, and GNU Radio, the most widely used open-source SDR framework, already implements it. Closing this gap would make ISAC experiments accessible to any lab with a software radio, not just this project.

1.3 Assumptions

This work assumes a single node transmits and receives its own reflection, treats targets as simple point reflectors, and assumes ideal synchronization for early testing. These are common simplifications in OFDM-radar research, useful for isolating the sensing problem from other sources of error, and will be revisited once implementation begins.

1.4 Scope

The project covers reusing GNU Radio's existing OFDM blocks and adding sensing on top of them, evaluated against KPIs such as BER, throughput, and range or Doppler error. Multiple antennas, non-OFDM waveforms, and multi-target tracking fall outside this scope for now, since each would add complexity that is better tackled once the single-antenna case works.

2. Review of Existing Work

Three sources anchor the technical direction of this project: two recent survey/tutorial papers on OFDM-based ISAC, and GNU Radio's own existing OFDM implementation, which defines the baseline this project extends. Each has a limitation that matters for this project specifically, even where the source itself is strong.

2.1 Q. Dai et al., "A Tutorial on MIMO-OFDM ISAC: From Far-Field to Near-Field" (IEEE Communications Surveys & Tutorials, 2026)

This 2026 tutorial provides the shared signal model this project builds on: the same information used to correct a received signal can also reveal a target's range and speed. Its analysis uses a Cramer-Rao-bound comparison of communication and sensing accuracy, built for multi-antenna, near-field systems. The limitation for this project is fit, not correctness: a GNU Radio and USRP setup is single-antenna and far-field, so the tutorial's results do not transfer directly and have to be scaled down.

2.2 M. Mirabella, P. D. Viesti, A. Davoli, G. M. Vitetta, "Deterministic Signal Processing Techniques for OFDM-Based Radar Sensing: An Overview" (IEEE Access, 2023)

This 2023 survey compares OFDM-radar processing techniques, including the reciprocal-filtering method adopted in Section 3, judging each on resolution, sidelobe behavior, and computational cost. It is thorough at the algorithm level but stops there: none of the techniques it covers are implemented in real SDR software, which is the limitation this project is built to address.

2.3 GNU Radio's Existing OFDM Implementation (gr-digital/ofdm_txrx.py, OFDM Transmitter/Receiver wiki page)

GNU Radio's OFDM code already handles synchronization, equalization, and demodulation, and computes a channel estimate as part of that process. The limitation is that this estimate is discarded once equalization is done. Nothing in the existing codebase exposes it or turns it into a range or Doppler estimate, which is the specific gap this project closes.

2.4 Additional Supporting Literature

Four more peer-reviewed papers were reviewed to round out this picture, each with its own limitation relative to this project:

- **Sit et al. (2011, SPACOMM).** An earlier joint radar-communication overview from the same research group behind the 2D-FFT algorithm this project uses. Like the algorithm itself at that stage, it stays theoretical and does not touch real hardware or software.
- **Liu et al. (2020, IEEE Trans. Commun.) and Zhang et al. (2022, IEEE Comm. Surveys and Tutorials).** Broad surveys that place this project within the wider joint communication-and-sensing research area. Their scope is broad rather than deep: neither focuses on OFDM specifically or on GNU Radio as a platform.
- **Baquero Barneto et al. (2019, IEEE TMTT).** A hardware demonstration of OFDM radar on real LTE and 5G NR waveforms. It proves the idea works on real signals, but the implementation is not open source and does not target GNU Radio.

Together, these confirm the chosen approach has both theoretical grounding and practical precedent, even though none of them close the specific gap this project targets: a tested, open-source GNU Radio block.

3. State-of-the-Art (SOTA) Position

The strongest existing approach for a single-antenna GNU Radio setup is reciprocal filtering with a 2D-FFT, introduced by Sturm and Wiesbeck (2011) and surveyed in Mirabella et al. (2023). It isolates what the channel did to the signal, then reads off range and speed from it.

This method is preferred over correlation-based alternatives because it is cheaper to compute, and over learning-based sensing because it needs no training data. Its resolution and ambiguity limits can be worked out with simple math before anything is built, which matters at this stage since nothing has been implemented yet. It is the method this project will implement in Milestone 2, and this SOTA position will be refined throughout subsequent milestones as implementation and testing progress.

4. Project Timeline

The plan below runs from the start of this milestone through final submission:

1. Foundation (this report)	Sep 15 to Sep 28, 2026	M1	Problem definition, literature review, SOTA positioning.
2. Baseline implementation	Sep 29 to Oct 19, 2026	M2	Study GNU Radio's OFDM code and build the sensing block.
3. Integration & validation	Oct 20 to Nov 9, 2026	M2/M3	Integrate and test the block, then validate it against KPIs.
4. Optimization & major contribution	Nov 10 to Nov 30, 2026	M3	Optimize the block and add a larger contribution (sidelobe suppression).
5. Finalization	Dec 1 to Dec 7, 2026	M3 / Final	Finalize documentation and submit for review.

5. References

[1] Q. Dai et al., "A Tutorial on MIMO-OFDM ISAC: From Far-Field to Near-Field," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 4319–4358, 2026, doi: 10.1109/COMST.2025.3650568.

[2] M. Mirabella, P. D. Viesti, A. Davoli, and G. M. Vitetta, "Deterministic Signal Processing Techniques for OFDM-Based Radar Sensing: An Overview," *IEEE Access*, vol. 11, pp. 68872–68889, 2023, doi: 10.1109/ACCESS.2023.3292937.

[3] C. Sturm and W. Wiesbeck, "Waveform Design and Signal Processing Aspects for Fusion of Wireless Communications and Radar Sensing," *Proceedings of the IEEE*, vol. 99, no. 7, pp. 1236–1259, 2011, doi: 10.1109/JPROC.2011.2131110.

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[5] F. Liu, C. Masouros, A. P. Petropulu, H. Griffiths, and L. Hanzo, "Joint Radar and Communication Design: Applications, State-of-the-Art, and the Road Ahead," *IEEE Transactions on Communications*, vol. 68, no. 6, pp. 3834–3862, 2020, doi: 10.1109/TCOMM.2020.2973976.

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[8] "OFDM Transmitter — GNU Radio," GNU Radio Wiki. https://wiki.gnuradio.org/index.php/OFDM_Transmitter

[9] GNU Radio, "gr-digital/python/digital/ofdm_txrx.py," GitHub. https://github.com/gnuradio/gnuradio/blob/main/gr-digital/python/digital/ofdm_txrx.py